

第四章 有限元方法收敛性理论.

Cea 引理. $\|u - u_h\|_{H^1} \leq C \inf_{v_h \in V_h} \|u - v_h\|_{H^1}$

§1. 插值理论

快速计算

整数 有界线性算子

引理 (Bramble - Hilbert) Ω 有界, Lip. $m \geq 1$ $\pi: H^m \rightarrow H^m \in 2CH^m \cdot H^m$

Poincaré 推广

$\pi q_k = q_k \quad \forall q_k \in P_{m-1}$ 则 $\exists C = C(m, \Omega)$

保持 $m-1$ 次多项式不变.

$$\|v - \pi v\|_{H^m} \leq C |v|_{H^m} \quad \forall v \in H^m.$$

证: 令 $\pi_m: H^m \rightarrow P_{m-1}$ 按 H^m 内积 正交投影

利用了 $\pi q_k = q_k$ 以及线性.

$$\pi_1: H^1 \rightarrow P_0$$

$$\pi_1 v = \frac{1}{|\Omega|} \int_{\Omega} v$$

$$\|v - \pi v\|_{H^m} = \|v - \pi_m v - \pi(v - \pi_m v)\|_{H^m} \leq C \|v - \pi_m v\|_{H^m}.$$

$$V = \{v \in H^m : \pi_m v = 0\}.$$

只需证 $\|v\|_{H^m} \leq C |v|_{H^m} \quad \forall v \in V$

反证:

$$\text{设 } \exists v_n \in V \quad \text{s.t.} \quad \|v_n\|_{H^m} = 1 \quad |v_n|_{H^m} \leq \frac{1}{n}$$

$$H^m \hookrightarrow H^{m-1} \quad \exists \text{ 子列 (记为 } \{v_n\}) \quad \text{s.t.} \quad v_n \xrightarrow{H^{m-1}} v$$

$$\{v_n\} \text{ 在 } H^m \text{ 中是 Cauchy 列.} \Rightarrow v_n \xrightarrow{H^m} v \in H^m.$$

$$\Rightarrow \|v\|_{H^m} = 1 \quad |v|_{H^m} = 0 \quad v \in V$$

$$v \in P_{m-1} \quad \pi_m v = 0 \quad \text{矛盾} \quad \square.$$

$$\text{例 } m=1, \quad \pi v = \frac{1}{|\Omega|} \int_{\Omega} v \quad \text{Poincaré} \quad \Omega_0 \quad \|v - v_{\Omega_0}\|_{L^2(\Omega)} \leq C \|\nabla v\|_{L^2(\Omega)}$$

$$m=1, \quad \pi v = \frac{1}{|\Omega|} \int_{\Omega} v \quad \text{Friedrichs}$$

$$\|v - \frac{1}{|\Omega|} \int_{\Omega} v\|_{H^1} \leq C |v|_{H^1} \quad \forall v \in H^1 \quad \text{在 } \mathbb{R}^n \text{ 上也对.}$$

$$\|v\|_{H^1} \leq C(|v|_{H^1} + |\int_{\Omega} v|) \quad |\Omega| > 0$$

Deny-Lions

引理 $\inf_{q_k \in P_{m-1}} \|v - q_k\|_{H^m} \leq C |v|_{H^m} \quad \forall v \in H^m.$

K 与 $\mathbb{R}$ 仿射等价: $\exists$ 可逆仿射变换 $x = F(x) = Bx + b: \mathbb{R} \rightarrow K.$

引理: K 与 $\mathbb{R}$ 仿射等价, $\varphi(x) = v(x) \quad m \geq 0$

则

$$(i) \quad |\varphi|_{H^m(K)} \leq C \|B\|^m |\det B|^{-\frac{1}{2}} |v|_{H^m(\mathbb{R})}$$

$$(ii) \quad |v|_{H^m(K)} \leq C \|B^{-1}\|^m |\det B|^{\frac{1}{2}} |\varphi|_{H^m(\mathbb{R})}$$

其中 $C = d^{\frac{m}{2}}$ $\|B\|^m = \sup_{0 \neq \xi \in \mathbb{R}^d} \frac{|B\xi|}{|\xi|}$

$(x_1, \dots, x_d)$

证: $\|\widehat{\varphi}\|_{H^m(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} |\widehat{\partial_{i_1}} \widehat{\partial_{i_2}} \dots \widehat{\partial_{i_m}} \widehat{\varphi}(x)|^2 dx$

$\|\varphi\|_{H^m(\mathbb{R}^d)}^2 = \int_{\mathbb{R}^d} |\partial_{j_1} \partial_{j_2} \dots \partial_{j_m} \varphi(x)|^2 dx$

$\widehat{\partial_{i_1}} \widehat{\varphi} = \frac{\partial \widehat{\varphi}}{\partial x_{i_1}} = \sum_j \frac{\partial \varphi}{\partial x_j} \frac{\partial x_j}{\partial x_{i_1}} = \sum_j b_{ji} \frac{\partial \varphi}{\partial x_j}$

$\widehat{\partial_{i_1}} \dots \widehat{\partial_{i_m}} \widehat{\varphi} = \sum_{j_1 \dots j_m \leq d} b_{j_1 i_1} \dots b_{j_m i_m} \partial_{j_1} \dots \partial_{j_m} \varphi$

$|\widehat{\partial_{i_1}} \dots \widehat{\partial_{i_m}} \widehat{\varphi}|^2 = \left| \sum_{j_1 \dots j_m \leq d} b_{j_1 i_1} \dots b_{j_m i_m} \partial_{j_1} \dots \partial_{j_m} \varphi \right|^2$

$\leq \left(\sum_{j_1 \dots j_m \leq d} b_{j_1 i_1} \dots b_{j_m i_m} \right) \left(\sum_{j_1 \dots j_m \leq d} |\partial_{j_1} \dots \partial_{j_m} \varphi|^2 \right)$

$= \left(\sum_{j_1} b_{j_1 i_1} \right) \dots \left(\sum_{j_m} b_{j_m i_m} \right) \left(\sum_{j_1 \dots j_m \leq d} |\partial_{j_1} \dots \partial_{j_m} \varphi|^2 \right)$

记 e_i 为 $\mathbb{I}_{d \times d}$ 的第 i 列。则:

$\leq |Be_{i_1}|^2 \dots |Be_{i_m}|^2 (\quad)$

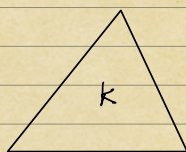
$\leq \|B\|^{2m} (\quad)$

引理: $\|B\| = \sup_{0 \neq \xi \in \mathbb{R}^d} \frac{|B\xi|}{|\xi|}$ $\|B\| \leq \frac{h_K}{\rho_K}$ $\|B^{-1}\| \leq \frac{h_K}{\rho_K}$ $|\det B| = \frac{|K|}{|K'|}$

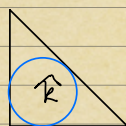
ρ_K K 上内切球直径

proof: $\|B\| = \sup_{|\xi|=1} \frac{|B\xi|}{1}$

$\widehat{\varphi} = \varphi_K$



$x - y = B\xi$



$\xi = x - y$

局部插值算子: 给定有限元 (K, P, N) 基 $\{\phi_1, \dots, \phi_n\}$ $N_i(\phi_j) = \delta_{ij}$

定义 $I_K \varphi \in P: I_K \varphi = N_1(\varphi) \phi_1 + \dots + N_n(\varphi) \phi_n$

$N_i(\varphi) = N_i(I_K \varphi)$ $I_K \varphi = \varphi$ if $\varphi \in P$

定理 1: $(K, \widehat{P}, \widehat{N})$ 是有限元, $m \geq 1$, $\ell \geq 0$, $\widehat{I} = I_K$ 有限元插值.

设 (i) $P_{m-1} \subset \widehat{P} \subset H^m(\mathbb{R}^d)$

(ii) $\widehat{N} \subset (C^\ell(\mathbb{R}^d))'$

(iii) $m - \frac{d}{2} > \ell$

$\Rightarrow k - \frac{d}{p} > 0$ $W^{k,p} \hookrightarrow C(\mathbb{R}^d)$ $H^m \hookrightarrow C^\ell$

则 $\|\varphi - \widehat{I} \varphi\|_{H^1(K)} \leq C \|\varphi\|_{H^m(K)} \quad \forall \varphi \in H^m(K) \quad 0 \leq i \leq m$

proof: 只需证 $\|\varphi - \hat{\Gamma}\varphi\|_{H^m(\mathbb{R})} \leq C \|\varphi\|_{H^m(\mathbb{R})}$

$\hat{\Gamma}\hat{q}_k = \hat{q}_k \quad \forall \hat{q}_k \in \hat{\mathcal{P}} \quad p_{m-1}$ 多项式成立.

$\hat{N}_j \in (C^0(\mathbb{R}))'$ 有界线性泛函连续性

$$\|\hat{\Gamma}\varphi\|_{H^m(\mathbb{R})} = \left\| \sum_{j=1}^n \hat{N}_j(\varphi) \hat{\phi}_j \right\|_{H^m(\mathbb{R})} \leq \sum_{j=1}^n |\hat{N}_j(\varphi)| \|\hat{\phi}_j\|_{H^m(\mathbb{R})}$$

$$\leq C \|\varphi\|_{C^0(\mathbb{R})} \leq C \|\varphi\|_{H^m(\mathbb{R})} \quad \square$$

(K, p, N) 与 $(\mathbb{R}, \hat{\mathcal{P}}, \hat{N})$

仿射插值等价. (i) K 与 $\mathbb{R}$ 仿射等价.

(ii) $\mathcal{P} = \{q_k : \hat{q}_k \in \hat{\mathcal{P}}\}$

(iii) $\hat{\Gamma}_K \varphi = \hat{\Gamma}_{\mathbb{R}} \hat{\varphi}$

定理 2: 设 (K, p, N) 与 $(\mathbb{R}, \hat{\mathcal{P}}, \hat{N})$ 仿射插值等价, 在定理 1 的条件下:

$$\|\varphi - \Gamma_K \varphi\|_{H^i(K)} \leq C \frac{h_K^{m-i}}{h_K} |\varphi|_{H^m(K)} \quad 0 \leq i \leq m.$$

其中 $C = C(K, m, d, \frac{h_K}{p_K})$

proof: $|\varphi - \Gamma_K \varphi|_{H^i(K)} \leq C \|B^{-1}\|^i |\det B|^{\frac{1}{d}} |\varphi - \hat{\Gamma}\varphi|_{H^i(\mathbb{R})}$

$$= C \|B^{-1}\|^i |\det B|^{\frac{1}{d}} |\varphi - \hat{\Gamma}\varphi|_{H^i(\mathbb{R})}$$

$$\leq C \|B^{-1}\|^i |\det B|^{\frac{1}{d}} \|\varphi\|_{H^m(\mathbb{R})}$$

$$\leq C \|B^{-1}\|^i \|B\|^m |\varphi|_{H^m(K)}$$

$$\leq C \left(\frac{h_K}{p_K}\right)^i \left(\frac{h_K}{p_K}\right)^m |\varphi|_{H^m(K)}$$

$$\leq C \left(\frac{h_K}{p_K}\right)^i h_K^{m-i} |\varphi|_{H^m(K)}$$

例 | Lagrange. 元.

线性 $m=2, 1 \quad k=0 \quad m-\frac{d}{2} > 0$

$d=1, \quad \|\varphi - \Gamma_K \varphi\|_{L^2} \leq C h_K \|\varphi\|_{H^1(K)}$

$d=1, 2, 3, \quad \|\varphi - \Gamma_K \varphi\|_{L^2(K)} \leq C h_K^{\frac{1}{2}} |\varphi|_{H^2(K)}$

二次元.

$d=1, \quad \|\varphi - \Gamma_K \varphi\|_{L^2} \leq C h_K |\varphi|_{H^1(K)}$

$d=1, 2, 3$ 简单略.....

设 $\{M_h\}$ 是 Ω 的一族剖分:

regular

正则: $\exists \mu > 0$ s.t. $\frac{h_K}{\rho_K} \leq \mu, \quad \forall K \in M_h.$

拟均匀: $\exists \nu > 0, \quad \frac{h}{h_K} \leq \nu, \quad h = \max_{K \in M_h} h_K.$
quasi-uniform.

定理 3: $(\mathbb{R}, \hat{P}, \hat{N})$ 是参考有限元, 满足定理 1.

$\{M_h\}$ 正则 $\forall K \in M_h, (K, P, N) \rightarrow (\mathbb{R}, \hat{P}, \hat{N})$ 仿射插值等价

$$I_h \varphi|_K = I_K(\varphi|_K)$$

$$\left(\sum_{K \in M_h} \|\varphi - I_h \varphi\|_{H^1(K)}^2 \right)^{\frac{1}{2}} \leq C h^{m-1} \|\varphi\|_{H^m(\Omega)}$$

$$\|\varphi - I_h \varphi\|_{L^2(\Omega)} \leq C h^m \|\varphi\|_{H^m(\Omega)}$$

$$\text{if } I_h \varphi \in H^1(\Omega) \quad \|\varphi - I_h \varphi\|_{H^1(\Omega)} \leq C h^{m-1} \|\varphi\|_{H^m(\Omega)}$$

逆估计: K 与 $\mathbb{R}$ 仿射等价, P 是 K 上有限维空间, $m \geq k \geq 0$ 则:

$$\|\varphi\|_{H^m(K)} \leq C h_K^{k-m} \|\varphi\|_{H^k(K)}$$

其中 $C = C(m, \mathbb{R}, \frac{h_K}{\rho_K}, d, \hat{P})$ $\forall \varphi \in \hat{P}$

proof: 不妨设 $k=0, \|\varphi\|_{H^m(K)} \leq C \|B^{-1}\|^m |\det B|^{\frac{1}{2}} \|\varphi\|_{H^m(\mathbb{R})}$

$$\leq C \|B^{-1}\|^m |\det B|^{\frac{1}{2}} \|\varphi\|_{H^m(\mathbb{R})}$$

$$\leq C \|B^{-1}\|^m |\det B|^{\frac{1}{2}} \|\varphi\|_{L^2(\mathbb{R})}$$

$$\leq C \|B^{-1}\|^m |\det B|^{\frac{1}{2}} |\det B|^{-\frac{1}{2}} \|\varphi\|_{L^2(K)}$$

$$\leq C \left(\frac{h_K}{\rho_K}\right)^m \|\varphi\|_{L^2(K)}$$

$$\leq C \left(\frac{h_K}{\rho_K}\right)^m h_K^{-m} \|\varphi\|_{L^2(K)}$$

整体逆估计需要拟均匀 且 $\varphi \in H^m(\Omega)$.

迹不等式

$$\|\varphi\|_{L^2(\partial\Omega)} \leq C \|\varphi\|_{L^2(\Omega)}^{\frac{1}{2}} \|\varphi\|_{H^1(\Omega)}^{\frac{1}{2}}$$

$$\|\varphi\|_{L^2(\partial K)} \leq C \|\varphi\|_{L^2(K)}^{\frac{1}{2}} \|\varphi\|_{H^1(K)}^{\frac{1}{2}}$$

引理: (局部迹不等式) K 与 $\mathbb{R}$ 仿射等价 则:

$$2ab \leq \varepsilon a^2 + \varepsilon^{-1} b^2$$

$$\forall \varphi \in H^1(K) \quad \|\varphi\|_{L^2(\partial K)} \leq C \left(h_K^{-\frac{1}{2}} \|\varphi\|_{L^2(K)} + \|\varphi\|_{L^2(K)}^{\frac{1}{2}} \|\nabla \varphi\|_{L^2(K)}^{\frac{1}{2}} \right) \leq C \left(h_K^{\frac{1}{2}} \|\varphi\|_{L^2(K)} + h_K^{\frac{1}{2}} \|\nabla \varphi\|_{L^2(K)} \right)$$

$$\|\varphi\|_{L^2(\partial\Omega)} \leq C \|\varphi\|_{L^2(\Omega)}^{\frac{1}{2}} \|\varphi\|_{H^1(\Omega)}^{\frac{1}{2}} \leq C \left(\|\varphi\|_{L^2(\Omega)} + \|\nabla \varphi\|_{L^2(\Omega)} \right)$$

proof: $\|\varphi\|_{L^2(\partial K)}^2 = \int_{\partial K} \varphi^2 dS \leq C \|B\|^{d-1} \int_{\partial \hat{K}} \hat{\varphi}^2 d\hat{S} = C \|B\|^{d-1} \|\hat{\varphi}\|_{L^2(\partial \hat{K})}^2$

$$dS = |x'(t)| = |B \hat{x}'(t)| = \|B\| d\hat{S}$$

$$\leq C \|B\|^{d-1} \|\hat{\varphi}\|_{L^2(\mathbb{R})} \|\hat{\varphi}\|_{H^1(\mathbb{R})}$$

$$\leq C \|B\|^{d-1} (\|\hat{\varphi}\|_{L^2(\mathbb{R})}^2 + \|\hat{\varphi}\|_{L^2(\mathbb{R})} \|\hat{\varphi}\|_{H^1(\mathbb{R})})$$

$$\leq C \|B\|^{d-1} |\det B|^{-1} (\|\varphi\|_{L^2(K)}^2 + \|B\| \|\varphi\|_{L^2(K)} \|\nabla \varphi\|_{L^2(K)})$$

$$\leq C \left(\frac{h_K}{\rho_K}\right)^{d-1} \frac{|R|}{|K|} (\|\varphi\|_{L^2(K)}^2 + \frac{h_K}{\rho_K} \|\varphi\|_{L^2(K)} \|\nabla \varphi\|_{L^2(K)})$$

$$\leq \left(\frac{h_K}{\rho_K}\right)^d \frac{1}{h_K} (\|\varphi\|_{L^2(K)}^2 + \frac{h_K}{\rho_K} \|\varphi\|_{L^2(K)} \|\nabla \varphi\|_{L^2(K)})$$

逆估计 $\|\nabla \varphi_h\|_{L^2(K)} \leq C h_K^{-1} \|\varphi_h\|_{L^2(K)}$ 有限维空间函数 φ_h .

$$\Rightarrow \|\varphi_h\|_{L^2(\partial K)} \leq \underline{C} h_K^{-\frac{1}{2}} \|\varphi_h\|_{L^2(K)}$$

与 p 相交与 h 无关

§2. H^1 估计

$$(P) \begin{cases} -\nabla \cdot (a \nabla u) + cu = f & \text{in } \Omega \\ u|_{\Gamma} = 0 \end{cases}$$

(VP) 求 $u \in H_0^1$ s.t.

$$a(u, \varphi) = (f, \varphi) \quad \forall \varphi \in H_0^1$$

(FEM) 求 $u_h \in V_h = \{ \varphi_h \in H_0^1 : \varphi_h|_K \in P_1 \quad \forall K \in \mathcal{T}_h \}$ s.t.

$$a(u_h, \varphi_h) = (f, \varphi_h) \quad \forall \varphi_h \in V_h$$

$$|a(u, \varphi)| \leq \beta \|u\| \|\varphi\|_1 \quad a(u, \varphi) \geq \alpha \|\varphi\|_1^2 \quad \forall \varphi \in H_0^1 \quad a(u, \varphi) = a(\varphi, u)$$

$\|\varphi\|_1 = \|\varphi\|_{H^1}$ $\|\varphi\|_1 = \|\varphi\|_{H^1}$

Cea 引理: $\|u - u_h\|_1 \leq \frac{\beta}{\alpha} \inf_{\varphi_h \in V_h} \|u - \varphi_h\|_1$

$$\|\varphi - I_h \varphi\|_1 \leq ch^2 \|\varphi\|_{H^2} \quad \|\varphi - I_h \varphi\|_1 \leq ch \|\varphi\|_{H^2}$$

定理 (H^1 估计) $\|u - u_h\|_1 \leq ch \|u\|_{H^2}$ (若 $u \in H^2$)

定理 (H^1 估计, 弱解)

proof: $\forall \varepsilon > 0 \quad \exists u_\varepsilon \in C^\infty \quad \text{s.t.} \quad \|u - u_\varepsilon\|_1 \leq \varepsilon$
取 $v_h = I_h u_\varepsilon$

$$\|u - u_h\|_1 \leq C \|u - I_h u\|_1 \leq C (\varepsilon + h \|u\|_{H^2})$$

$$\lim_{h \rightarrow 0} \|u - u_h\|_1 \leq C\varepsilon$$

$$\lim_{h \rightarrow 0} \|u - u_h\|_1 = 0$$

对偶论证:

$$(CP) \quad \begin{cases} -\nabla \cdot (a \nabla w) + cw = u - u_h & \text{in } \Omega \\ w|_{\Gamma} = 0 \end{cases}$$

$$a(w, \varphi) = (u - u_h, \varphi)$$

$$a(\varphi, w) = (\varphi, u - u_h) \quad \forall \varphi \in H_0^1$$

Ω 凸区域 (需要正则性, 弱解就是强解)

$$\|w\|_{H^2} \leq C (\|u - u_h\| + \|w\|)$$

$$\alpha \|w\|_1 \leq a(w, w) = (w, u - u_h) \leq \|w\| \|u - u_h\|$$

$$\Rightarrow \|w\| \leq C \|u - u_h\|$$

$$\leq C \|u - u_h\|$$

$$\|u - u_h\|^2 \leq a(u - u_h, w)$$

$$= a(u - u_h, w - I_h w)$$

$$\leq \beta \|u - u_h\|_1 \|w - I_h w\|_1$$

$$\leq C \|u - u_h\|_1 h \|w\|_{H^2}$$

$$\leq C \|u - u_h\|_1 h \|u - u_h\|$$

$$\Rightarrow \|u - u_h\| \leq Ch \|u - u_h\|_1 \leq Ch^2 \|u\|_{H^2}$$

作业: (P) $\begin{cases} u|_{\Gamma} = g \end{cases}$ 非齐次

$$(VP) \quad \begin{cases} \exists u \in H^1 & u|_{\Gamma} = g & \text{s.t.} & a(u, \varphi) = (f, \varphi) \quad \forall \varphi \in H_0^1 \\ \exists u_h \in V_h & u_h|_{\Gamma} = g_h & \text{s.t.} & a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h^0 \end{cases}$$